

CJ Geier

Evaluating the Effect of Play Action on Defensive Disruption and Offensive Efficiency

The main question I want to answer with this project is why play-action passing has seen such a massive resurgence across the NFL, specifically by looking at how it disrupts defensive metrics and drives offensive efficiency. While earlier football minds established that play-action was an important part of a balanced offensive attack, past work mostly relied on basic box-score statistics. Those traditional numbers miss what defenders are actually doing in the first second of the play, ignoring tracking metrics like get-off speed, pressure rates, and Expected Points Added (EPA) (NFL Big Data Bowl, 2024). Using the latest data from the NFL Big Data Bowl, I want to bridge that gap by measuring how run fakes directly impact the defense. I also want to see if running a heavy play-action offense creates positive effects for an offense's run game. The expected outputs for this project will be a formal research paper, a reproducible GitHub repository, and an interactive dashboard. For my methodology, I will use situational matching in R to control for down and distance, ensuring that play-action's success isn't just a byproduct of early down play-calling bias. The intellectual merit of this project will be connecting tracking data with overall offensive value to explain why play-action is such a vital part of modern passing attacks. It will also help the average fan understand the value of running play-action, and how it affects all facets of a team's offense.

I already have a strong foundation in most of the core skills needed for this project through both my coursework and hands-on work on campus. Classes like Advanced Descriptive Methods and Business Analytics have given me a lot of experience cleaning messy data, running

statistical models, and working in R. Through my work with Denison Strength and Conditioning, I have a lot of experience with extremely large datasets, and have successfully built clean dashboards for our coaches. I have also worked with real football data before with the Denison football team, where I charted and analyzed defensive metrics like pass rush win rates and double team rates. I do, however, have some areas for growth and learning this semester. While I know my way around football data and building interactive tools, I haven't worked directly with NFL tracking coordinates yet, and I will need to learn some more advanced methods to really get into depth with this data. To learn these methods, I will look into online tutorials, continue to spend time outside of class reading sports analytics books, and ask faculty members questions if I am really stumped.

As previously mentioned, in football analytics it is widely accepted that play-action passing boosts offensive efficiency, with studies showing noticeable spikes in yards per attempt and success rates regardless of whether an offense establishes the run game first (Sharp, 2019). While coaches and analysts know that play-action works and teams are using it more every year, there is very little to no research explaining why it works. Most existing analysis stops at simple statistical benchmarks, such as looking at box scores or EPA. There isn't really any current research measuring how run fakes disrupt defensive fronts at the snap. Metrics such as pressure rate and time to pressure will give a unique view of this question that doesn't seem to have been done before. My project fits into this gap by using data from the NFL Big Data Bowl to unpack that unknown why (NFL Big Data Bowl, 2024). By using the metrics listed above, I can measure the true physical deception of play fakes and show how important this deception is in offensive efficiency.

Phase:	Timeframe:	Completion Date
Initiation	Weeks 1–2	September 11th
Planning	Weeks 3–5	October 2nd
Execution	Weeks 6–12	November 20th
Closure	Weeks 13–15	December 11th
Monitoring	Semester Long	Weekly

To make sure this project stays on track and gets completed on time, I have organized my semester around the research life cycle shown in the table above. During the initiation phase, I got an early jump on things by defining my scope, setting up my GitHub repository for file organization and version control, finding my dataset, and finishing this project proposal. Moving into planning and early execution, my focus will be on thoroughly exploring the dataset, as it is extremely large, running some descriptive analysis, and matching the data to the situations in the game to control for down and distance. A big lesson I took away from DA 301 was the importance of staying ahead of my schedule, so my goal is to finish major coding and modeling tasks about a week before their deadlines. This will help give me some breathing room to review my work and get constructive feedback from faculty, with enough time to correct any mistakes. I will also consistently use GitHub pushes to make sure I have good version control, and will keep all my files organized, so my project is easily reproducible. In the final two weeks during the closure phase, I want to leave myself plenty of time to dedicate all my attention to designing my poster and practicing my presentation so I can finish the semester with a bang!

References

National Football League. (2024). *NFL Big Data Bowl 2024*. Kaggle.

<https://www.kaggle.com/competitions/nfl-big-data-bowl-2024>

Sharp, W. (2019, July 23). *The best time to utilize play action*. Sharp Football Analysis.

<https://www.sharpfootballanalysis.com/analysis/the-best-time-to-utilize-play-action/>